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Summary

Physicist developing machine-learning methods that make simulation and inference in particle physics faster, more accurate and transferable, and carrying them into the software that experiments use: generative surrogates for highly granular calorimeters, transfer between detector geometries, and benchmarks that test models where they will be used. Nine years of building, validating and deploying research software in collaborations from a handful of people to ATLAS; supervision of PhD, master's and undergraduate students; lead applicant of a DFG Emmy Noether proposal for an independent research group.

Academic Posts

January 2024
to present
Deutsches Elektronen-Synchrotron (DESY), Hamburg

Postdoctoral researcher – FTX (Future Experiments) – SFT (Software Division)
Generative machine learning for detector simulation at future Higgs factories and the HL-LHC, funded through the BMBF ErUM-Data project KISS. Responsible for the group's software products: packaging, continuous integration, model export for C++ inference and deployment on the Helmholtz Model Zoo. Development and training on DESY's Maxwell GPU cluster (SLURM).

- Led the development of CaloClouds 3, a hybrid of normalising flows and a distilled diffusion model that generates showers as geometry-independent point clouds: the fastest accurate photon-shower model for a highly granular calorimeter, about 100 times faster than Geant4 at inference; open source and a flagship model on the Helmholtz Model Zoo (publication 1).
- Collaborated on AllShowers, the first single generator for showers of 12 particle species, built from continuous normalising flows and a transformer with masked multi-head attention (publication 3).
- Contributed to the first full physics benchmark of calorimeter surrogates: the models were integrated into the ILD reconstruction chain (C++ inference via TorchScript and LibTorch, in Key4hep and DD4hep through DDML), an integration led by T. Madlener and A. Korol, and compared with Geant4 on analysis-level observables (publication 2).
- Collaborated on multi-geometry pre-training: a shower generator pre-trained on a synthetic prior of many simple geometries transfers, zero-shot or with fine-tuning, to unseen detector geometries and materials such as the ALLEGRO concept (publication 4).
- Co-supervise work on a foundation model that simulates both jets and calorimeter showers; investigating generative simulation of extensive air showers with the CORSIKA 8 team, for which I released the open-source package subshowers.

September 2021
to December 2023
Czech Technical University in Prague (CTU)

Postdoctoral researcher – ATLAS experiment
Validation of AtlFast3, the ATLAS fast simulation, and development of analysis software for inclusive jet cross sections; production on the CERN grid, with additional resources from the CTU cluster. ATLAS author from 1 March 2023 to 1 January 2026.

- Built and operated the validation pipeline for candidate FastCaloSim models (HTCondor and DAGMan on the grid), keeping pace with a continuous stream of new models; worked with analysis groups across the collaboration to agree benchmarks; coordinated computing resources and manual review contributed by several institutes; wrote the documentation and CI tests. This contributed to record uptake of fast simulation, over 60% in Run 2.
- Extended the FastCaloSim inference code in Athena, a codebase of about 7 million lines of C++, to select the inference back-end (ONNX Runtime or SOFIE) at runtime from the model format; unit and regression tests in CI.
- Developed a columnar (ROOT RDataFrame) approach to jet and dijet cross-section calculations over the full set of jet triggers, with data handled through Rucio; replicated a published Chebyshev-polynomial fit of the total jet cross section with several optimisers to establish the conditions for a stable fit.

Member of ATLAS during the period recognised by the 2025 Breakthrough Prize in

Fundamental Physics.

Education

September 2017 to September 2021
University of Southampton.
Degree awarded February 2022.

PhD in Physics – machine learning for high-energy physics
Thesis: *Extracting jet signals of New Higgs Physics: from traditional analysis to machine learning*. Novel jet-clustering and flavour-tagging algorithms for particle physics, developed with the CMS group and a supervisory team spanning phenomenology (Prof. Stefano Moretti), computer science (Prof. Srinandan Dasmahapatra) and experiment (Prof. Claire Shepherd-Themistocleous). Funded by STFC and EPSRC. Code released openly (SPECTRALJET, JETTOOLS).

- Next Generation Computational Modelling (NGCM) doctoral training centre (2017–2018): a year of formal training in scientific programming (PYTHON, C++), numerical methods, statistical analysis, simulation and modelling, HPC and GPU programming (OPENMP, CUDA); alongside the NExT Institute's particle-theory programme.
- Secondment to Cadi Ayyad University, Morocco (2019; European Commission RISE funding), supporting extended Higgs-sector studies: performed the computations behind publication 10, interfacing FORTRAN 77 and C++ from PYTHON, containerising a legacy stack and refactoring the FORTRAN toolchain for about 30% faster inference.

September 2012 to June 2017
University of Edinburgh

MPhys in Theoretical Physics, first class
First-class MPhys (integrated master's degree) in Theoretical Physics; Year 2 Certificate of Merit for Theoretical Physics. Summer research projects with TU Munich (2015, production rates of stoponium; funded by TUM) and the Hyper-Kamiokande collaboration (2014, simulation of photomultipliers).

Current and Future Research

My current work builds generative surrogates for highly granular calorimeters, tests them inside full reconstruction chains on the observables analyses use, and pre-trains them so that they transfer between detector geometries. Over the coming years I plan to build an independent group around three strands:

- **Zero-shot and data-driven learning for detector simulation:** models that take the detector geometry as runtime context, so that one model serves many detectors and can be trained on beam-test data to predict the real detector, a novel alternative to the Monte Carlo process; first application with the CMS high-granularity calorimeter.
- **Community benchmarks for generative models in high-energy physics:** an accessible pipeline that compares models across the field on downstream, analysis-level metrics and presents the results as speed–accuracy Pareto fronts.
- **Data-driven learning for extensive air showers:** transferring fast-simulation techniques to air showers, where simulation statistics limit analyses, and using data-driven training on Pierre Auger Observatory data to study the muon puzzle; groundwork already released as SUBSHOWERS.

The first two strands serve collider physics and the third connects it to astroparticle physics; the same pattern of fast differentiable surrogates with per-instance adaptation extends to inverse problems beyond particle physics, including in-vivo verification of proton therapy.

Research Grants and Contracts

DFG Emmy Noether Programme (lead applicant)
Applicant and lead author of a proposal for an independent junior research group on generative machine learning, zero-shot generation and data-driven training as alternatives to Monte Carlo detector simulation, with interdisciplinary objectives in extensive air showers. Six-year programme with collaborators at six institutes in Germany, the Czech Republic and the UK. Submitted in early 2026 after passing DESY's internal review; under review.

KISS (BMBF ErUM-Data), 2024–present
"Künstliche Intelligenz zur schnellen Simulation von wissenschaftlichen Daten" (AI for the fast simulation of scientific data), a joint project funded by the German Federal Ministry of Education and Research, coordinated by Prof. Gregor Kasieczka (Universität Hamburg) and spanning particle, nuclear and astroparticle physics and astronomy. My DESY position has been funded through KISS; CALOCLOUDS 3, ALLSHOWERS and the full-physics benchmark contribute directly to its aim of accurate, resource-efficient simulation, and I presented CALOCLOUDS 3 at the KISS meeting in Munich (2025).

Earlier funding	PhD studentship, University of Southampton (2017–2021), funded by STFC and EPSRC, including the EPSRC-funded NGCM doctoral training centre and the STFC-funded NExT Institute. European Commission RISE staff-exchange funding for the secondment to Cadi Ayyad University (2019), which led to publication 10. Summer research funding from TU Munich (2015).
Computing resources	GPU training and deployment on DESY's Maxwell cluster (2024–present); large-scale production on the CERN grid with additional resources from the CTU cluster (2021–2023).
Supporting others' funding	Feedback on and proofreading of Prof. Zdeněk Hubáček's section on the CTU Prague ATLAS group for the successful FORTE application ("Fundamental constituents of matter through frontier technologies", a five-institution excellence project under the Czech Operational Programme Jan Amos Komenský, 2024–2028). Support for students' individual applications for third-party funding. With DESY's Early Career Scientist Board, organising an internal seminar series in which grant holders explain how they developed and wrote their proposals.

Student Supervision

PhD students (DESY, co-supervised)	**Anatolii Korol** (graduated March 2026). Projects included CaloClouds 3 and the first full physics benchmark for highly granular calorimeter surrogates (publications 1 and 2). **Diptarko Choudhury** (since September 2026). Faster inference for AllShowers and a more efficient data and evaluation toolkit.
Research projects (co-supervised)	**Eric Schaefer** (DESY and HAW Hamburg, in progress): *Foundation Models for Jets and Showers in Future Detectors*, a model that simulates across the physical modalities of jets and calorimeter showers; the project also builds knowledge exchange between the two institutes.
Summer and project students (DESY)	**Jemma Bagg** (2026) replaced the longitudinal-profile flow of CaloClouds 3 with the continuous normalising flow used in AllShowers: faster at inference, fewer parameters and more accurate. **Aysu Kocaalp** (2025) inverted the normalising flow for photon-shower profiles into a reconstruction tool and compared it with standard reconstruction. **Simon Edgar Pijahn** (2024) built a classifier that separates generative from full Monte Carlo simulation and ranks generative models by their distance from Monte Carlo. All three have gone on to PhD or master's study.
Assisting supervision	DESY and Universität Hamburg (2024–present): active in both groups (software for future detectors, generative detector simulation, foundation models), helping students interpret their findings in a broader context and form interdisciplinary connections, such as links to CORSIKA 8. CTU Prague and ATLAS (2021–2023): batch systems (SLURM, HTCondor), core software (ROOT, Geant4), jet-substructure metrics; in the ATLAS fast-simulation team, the architecture of the inference pipeline, bookkeeping, validation pipelines and the physics behind the validation of new generative models. DMLab exchange (CNRS Helmholtz Dark Matter Lab): advising Dr Dirk Zerwas's students at Université Paris-Saclay on fast inference.

Teaching

Undergraduate teaching	University of Southampton (2019 and 2020; classes of about 30 students). *Simulation and Modelling*, demonstrator and assessor: led workshop sessions and assessed coursework in two modes, a marked submission followed by a short oral assessment of each student's intent and understanding. *Introduction to Machine Learning*, demonstrator: led workshops on clean code and the core principles of machine learning; marked submitted work with detailed feedback. I marked both examinations and coursework.
Earlier teaching	Mathematics tutor at Standard Grade and Higher level, Edinburgh Tutors (2013–2014). Designed and delivered week-long workshops in Java programming and electronics for 8–12-year-olds as a self-employed tutor (2014–2018): programme design, teaching materials, marketing and delivery.
Postgraduate training and seminars	*Fundamentals of neural networks*, JČF workshop, Bílý Potok (2023): a bottom-up, mathematical explanation of neurons, networks and backpropagation with live audience participation. *Design patterns for particle-physics software*, JČF workshop (2022; proceedings paper). Invited keynote on LLMs for users, 8th Round Table on Deep Learning, DESY (2025). Invited seminar on the academic career path, DESY Fellows Day (2026). Beamline for Schools, CERN (2025 and 2026): in-person research assistant for

	several visiting teams each year, teaching data extraction, curation and analysis on a real experiment.
Programmes designed and run	ATLAS introductory days at CERN (ECSB representative, 2022–2023): programme design driven by feedback surveys, speaker invitations and worldwide advertising. DESY summer student programme, FTX topical group (2024–2026): introductions, final presentations, timetable, evaluations and reporting. Joint ML Journal Club (founder and coordinator, 2025–present): DESY, Universität Hamburg and HAW Hamburg, with industry speakers; about ten participants in a typical session and close to 40 for the most popular.
Feedback	Feedback on the 2023 JČF neural-networks seminar was very positive, with students relieved to have demystified the topic; the LLM keynote drew requests for more extended training; the Fellows Day seminar prompted an extended discussion between senior staff and the fellows. All three summer students I have supervised completed their projects; one is now part of CaloClouds 3 and another became a metric we use to rank generative models.
Teaching interests	Detector physics, programming and numerical methods, statistics and data analysis, simulation methods for physics, and machine learning for physics. At master's level I would like to develop modules on *Generative Models for Surrogate Simulation* and *Foundation Models in Physics*, each combining lectures with a project that takes a model from prototype to tested, deployable research software.

Publications

In line with DORA, I state my own contribution to each output; author lists in my field are usually alphabetical or collaboration-wide. ORCID 0000-0002-9710-2980; INSPIRE-HEP inspirehep.net/authors/2636704.

Peer-reviewed journal articles and preprints

March 2026
JINST 21 (2026) P03018

1. CaloClouds3: Ultra-Fast Geometry-Independent Highly-Granular Calorimeter Simulation
T. Buss, H. Day-Hall, F. Gaede, G. Kasieczka, K. Krüger, A. Korol, T. Madlener, P. McKeown, M. Mozzanica, L. Valente. arXiv:2511.01460.
https://doi.org/10.1088/1748-0221/21/03/P03018
Contribution: I led the development, from architecture through validation to deployment in the reconstruction chain, and co-supervised the PhD student working on it (A. Korol). The fastest accurate photon-shower model for a highly granular calorimeter, about 100 times faster than Geant4 at inference; a flagship model on the Helmholtz Model Zoo and a DESY research highlight at ICCS 2026.

2026
Physical Review D

2. A First Full Physics Benchmark for Highly Granular Calorimeter Surrogates
T. Buss, H. Day-Hall, F. Gaede, G. Kasieczka, K. Krüger, A. Korol, T. Madlener, P. McKeown. arXiv:2511.17293.
https://doi.org/10.1103/fn99-hq7q
Contribution: I co-supervised the PhD work behind this study (A. Korol) and contributed to the integration of the surrogates into the ILD reconstruction chain with C++ inference, which T. Madlener and A. Korol led; we compared the surrogates with Geant4 on analysis-level observables. The first test of calorimeter surrogates on full-physics, end-use metrics.

2026
SciPost Physics 21, 076

3. AllShowers: One model for all calorimeter showers
T. Buss, H. Day-Hall, F. Gaede, G. Kasieczka, K. Krüger. arXiv:2601.11716.
https://doi.org/10.21468/SciPostPhys.21.3.076
Contribution: Co-developer of the model. The first single generator for showers of 12 particle species, using continuous normalising flows and a transformer with masked multi-head attention.

August 2026
Submitted to JINST

4. Transferable Fast Calorimeter Shower Generation via Multi-Geometry Pre-training
T. Buss, H. Day-Hall, F. Gaede, G. Kasieczka, K. Krüger, L. Valente. arXiv:2608.18233.
https://doi.org/10.48550/arXiv.2608.18233
Contribution: Collaborator on the transfer-learning study. Shows that shower generators transfer between geometries and materials, and finds the break-even points of zero-shot transfer and fine-tuning against training from scratch.

February 2022
JHEP 02 (2022) 165

5. Spectral Clustering for Jet Physics
S. Dasmahapatra, H.A. Day-Hall, B. Ford, S. Moretti, C.H. Shepherd-Themistocleous.
https://doi.org/10.1007/JHEP02(2022)165
Contribution: The core of my PhD. I adapted spectral clustering to be infrared- and collinear-safe, demonstrated its utility in a two-Higgs-doublet-model scenario, and wrote and released the open-source package SpectralJet.

March 2025
European Physical Journal C

6. Software and computing for Run 3 of the ATLAS experiment at the LHC
ATLAS Collaboration.
https://doi.org/10.1140/epjc/s10052-024-13701-w
Contribution: Validation of AtlFast3.

October 2024
Phys. Rev. D 110 (2024) 072019

7. Measurements of jet cross-section ratios in 13 TeV proton-proton collisions with ATLAS
ATLAS Collaboration.
https://doi.org/10.1103/PhysRevD.110.072019
Contribution: Columnar analysis development for total jet cross sections.

April 2023
European Physical Journal C

8. Fat b-Jet Analyses Using Old and New Clustering Algorithms in New Higgs Boson Searches at the LHC
A. Chakraborty, S. Dasmahapatra, H. Day-Hall, B. Ford, S. Jain, S. Moretti, E. Olaiya, C. Shepherd-Themistocleous.
https://doi.org/10.1140/epjc/s10052-023-11537-4
Contribution: I helped reuse the existing pipelines and datasets, and explored the output data to understand the impact of the choices made in jet formation and to refine them.

April 2022
European Physical Journal C

9. Revisiting jet clustering algorithms for new Higgs Boson searches in hadronic final states
A. Chakraborty, S. Dasmahapatra, H. Day-Hall, B. Ford, S. Jain, S. Moretti, E. Olaiya, C. Shepherd-Themistocleous.
https://doi.org/10.1140/epjc/s10052-022-10314-z
Contribution: I generated the input datasets and helped build the analysis pipeline.

November 2020
Phys. Lett. B 810 (2020) 135819

10. Mapping $pp \to A \to ZH \to \ell^+\ell^- b\bar{b}$ and $pp \to H \to ZA \to \ell^+\ell^- b\bar{b}$ Current and Future Searches onto 2HDM Parameter Spaces
R. Benbrik, H. Day-Hall, S. Moretti, S. Semlali.
https://doi.org/10.1016/j.physletb.2020.135819
Contribution: Computational analysis during my secondment to Cadi Ayyad University: I performed all the computations and produced the plots, interfacing Fortran 77 and C++ from Python and building container stacks so that the required libraries could run without administrative privileges, and profiled and refactored the Fortran toolchain for about 30% faster inference.

Collaboration authorship

As a member of the ATLAS Collaboration I am a co-author of its papers from my authorship period (1 March 2023 to 1 January 2026); publications 6 and 7 are those with a specific contribution of mine.

Proceedings and thesis

September 2026
Conference proceedings, LHCP 2026

Point-cloud generative models for fast calorimeter simulation across particles and geometries
T. Buss, H. Day-Hall, F. Gaede, G. Kasieczka, K. Krüger, A. Korol, T. Madlener, P. McKeown, M. Mozzanica, L. Valente. arXiv:2609.30403.
https://arxiv.org/abs/2609.30403

September 2026
Conference proceedings

Jevons' Paradox and Fast Generative Simulation for HEP: Why Realistic Benchmarking is Essential
T. Buss, H. Day-Hall, F. Gaede, G. Kasieczka, K. Krüger, A. Korol, T. Madlener, P. McKeown, M. Mozzanica, L. Valente. arXiv:2609.24306.
https://arxiv.org/abs/2609.24306

June 2022
Proceedings of the JČF workshop, Bílý Potok

Design Patterns in Data Science
H. Day-Hall.
https://indico.fjfi.cvut.cz/event/200/attachments/1238/1876/book_of_proceedings.pdf

2022
PhD thesis, University of Southampton

Extracting jet signals of New Higgs Physics: from traditional analysis to machine learning
H. Day-Hall. Supervisors: S. Moretti, S. Dasmahapatra, C. Shepherd-Themistocleous.
https://eprints.soton.ac.uk/473633/

Research software and other outputs

Research software

CaloClouds 3, open source, flagship model on the Helmholtz Model Zoo: github.com/FLC-QU-hep/CaloClouds-3. AllShowers, packaged with pyproject, full continuous integration: github.com/FLC-QU-hep/AllShowers. subshowers, tools that split CORSIKA 8 air showers into sub-showers and compute a ground array's response, the data layer for my air-shower programme: github.com/HenryDayHall/subshowers. spectraljet and jetTools, my PhD jet-clustering and tagging code: github.com/HenryDayHall.

Interactive tools — Physics Playground, an interactive, bilingual (German and English) neural-network classification demonstration with realistic physics data, built for DESY's City of Science: physics-playground.day-hall.eu.

Invited Talks

October 2026 — **12th annual Matter and Technologies meeting of the Helmholtz Association**
Dresden — Invited overview "Detector Simulation and Control: Fast generative ML and agentic AI": DESY's and GSI's work on AI for accelerator and detector control (reinforcement learning and Bayesian optimisation for beam steering at ARES; an LLM-based assistant for detector control at the GSI Super-FRS; an agentic system in preparation for PETRA IV) alongside our fast generative detector simulation and transfer learning.

June 2026 — **DESY Fellows Day 2026**
DESY — Invited seminar on the academic career path: the dynamics of an academic career across European countries and five key non-research steps for an early career, with concrete, prioritised advice.

November 2025 — **8th Round Table on Deep Learning**
DESY — Invited keynote on LLMs for users: how LLMs work, their key flaws and the most significant developments, and the options open to users from online interfaces to APIs and terminal agents.

May 2021 — **Seminar, Prof. Mihoko M. Nojiri's group**
KEK, Tokyo — Invited seminar on the use of spectral clustering for jet formation.

December 2014 — **TEDx talk**
University of Edinburgh — Invited TEDx talk on the accuracy of Wikipedia and how to use it on unfamiliar topics; youtube.com/watch?v=-bGu4zKsbsE.

Other Selected Talks and Posters

July 2026 — **26th International Conference on Computational Science (ICCS 2026)**
DESY, Hamburg — Talk "Fast Generative Simulation for HEP" on the group's methods and benchmarking; CaloClouds 3 presented as a DESY research highlight.

September 2025 — **AI for Science**
Ljubljana — Talk on CaloClouds 3.

September 2025 — **ACAT 2025 (Advanced Computing and Analysis Techniques in Physics)**
Hamburg — Poster on CaloClouds 3; chair of the session on ML optimisation and evaluation.

March 2025 — **KISS collaboration meeting**
Munich — Talk on the development of CaloClouds 3.

June 2022 and June 2023 — **JČF workshops**
Bílý Potok — Seminars on design patterns for particle-physics software (2022) and on the fundamentals of neural networks (2023); see Teaching.

February 2023 — **RDataFrame on xAOD with friends**
CERN — Talk on the challenges of running ATLAS analysis in columnar format, presenting a pre-processing solution built on RDataFrame and friend trees.

October 2022 — **Efficient simulations on GPU hardware**
ETH Zürich — Talk on architecture and library considerations for optimally using GPU resources in GAN-based detector simulation.

June and October 2022 — **ATLAS weeks**
CERN and University of Lisbon — Posters on the composition of FastCaloSim and on the development and findings of its validation toolkit.

July 2019 — **1st Mediterranean Conference in Higgs Physics**
Tangier, Morocco — Talk on tools for determining parameter spaces, focused on container stacks in environments without administrative privileges.

Leadership, Service and Community

Early Career Scientist Boards — DESY FH board (elected representative, 2024–present): web content; communications and status updates from DESY's early-career researchers; the board's inbox and anonymous channel; design, running and reporting of large-scale surveys; regular communication with the physics directorate; the board's long-term strategy, presented each year at the physics retreat. ATLAS board (member, 2022–2023): web content, inbox and anonymous communications; team-building events; design and organisation of early-career workshops and training; surveys; regular reporting to the Physics Coordinators; coordinator of the ATLAS peer-mentoring programme.

Seminars and journal club Coordinator of the FTX internal seminar series, DESY (2024–present): balancing topics across the division's physics and engineering subgroups, inviting and timetabling speakers, technical support. Founder and coordinator of the Joint ML Journal Club (2025–present) for DESY, Universität Hamburg and HAW Hamburg, with industry speakers and sessions devoted to building collaboration between the institutes.

Conferences and workshops ACAT 2025, Hamburg: organisational support and chair of the session on ML optimisation and evaluation. ATLAS Standard Model Workshop 2023, CTU Prague: organisational support (budget, timetable, catering) and technical support in sessions. ATLAS introductory days at CERN (2022–2023) and the DESY summer student programme (2024–2026): see Teaching.

Peer review and collaborations Reviewer for Physical Review D. Collaborations: KISS (BMBF ErUM-Data, from 2024); ATLAS (2021–2023, with authorship from March 2023 to January 2026); ongoing work with the CORSIKA 8 team on air-shower simulation; an exchange with DMLab, the CNRS Helmholtz Dark Matter Lab, in which I advise Dr Dirk Zerwas's students at Université Paris-Saclay on fast inference and he advises on my fast-simulation work at DESY.

Developing others Coffee mornings with the ATLAS Physics Coordinators (ECSB, 2023): drop-in sessions run in person and by video conference, which I advertised, set up and moderated; they connected people with research teams and resources and showed the collaboration where its resources had gaps. Practical mental-health workshop (ATLAS ECSB, 2023): practical skills for professional resilience, well received across early-career researchers and permanent staff, with strong demand for further sessions. Internal seminar series on successful grant applications (DESY ECSB, in organisation).

Equality, diversity and inclusion Participant in DESY's diversity round tables; helped organise its queer film nights; through the DESY and ATLAS Early Career Scientist Boards, public representation of the trans* community in physics.

Public Engagement

2025 and 2026, CERN — **Beamline for Schools**
In-person experimental research assistant for several visiting high-school teams each year: Linux data-recording computers, data extraction over the network from the command line, automation with shell scripts, and statistical analysis in PYTHON, comparing the data with simulation and developing physics-based explanations for deviations.

June 2024, DESY — **DESY City of Science**
Designed a stall on how machine-learning classification is used in particle physics, from the detectors and their data to the fundamentals of neural networks, with an interactive, bilingual demonstration on realistic physics data (physics-playground.day-hall.eu).

March 2019, University of Southampton — **Southampton Science and Engineering Festival**
Live bench-top experiments and interactive performances demonstrating the concepts of a particle collider to primary-school children.

September 2018 to February 2019, DataKind UK — **Volunteer data analyst, Shelter**
Starting from a DataKind data dive, analysed data for the housing charity Shelter, transferring methods from high-energy physics.

Prizes and Recognition

2025 Breakthrough Prize in Fundamental Physics Shared, as an ATLAS co-author, in the prize awarded to the LHC collaborations for their Run 2 results. In ATLAS I maintained the fast-simulation back end, maintained and ran the validation systems, and contributed to the communication campaign that accompanied the record uptake of fast simulation.

Recognition of CaloClouds 3 Selected by the Helmholtz Model Zoo as a flagship model for fast calorimeter simulation; presented as a DESY research highlight at ICCS 2026, hosted by DESY in Hamburg.

University prizes Year 2 Certificate of Merit for Theoretical Physics, University of Edinburgh.

Professional Development

Schools and training NGCM doctoral training centre and NExT Institute, Southampton (2017–2018). 48th BUSSTEP, Oxford (November 2018). International School of Subnuclear Physics, Erice (June 2018). UCL summer school in AI and machine learning (July 2018). GRADnet winter school on leadership and team structuring (February 2018). Mental health first aid training, DESY (2024).

Continuing development	Before changing core ATLAS C++ code I studied software architecture so that my design would support future development of an extensive codebase. I keep current by moving into new areas: transformers and foundation models, LLM-assisted development, and air-shower simulation.

Computational Skills

Programming languages	*Experienced in:* Python, C++. *Also worked in:* C, Fortran 77, Bash, Java, T-SQL, LaTeX.
Development tools	git, with CI on GitHub and GitLab; unit and regression testing; packaging with pyproject, pip and conda; Docker and Apptainer; debuggers and profilers for Python and C++; LLM-assisted development.
HPC and distributed computing	SLURM (Maxwell), HTCondor and DAGMan, the CERN grid and Rucio; GPU training on clusters; OpenMP and CUDA.
Physics and ML software	ROOT/RDataFrame, Geant4, Key4hep, DD4hep, DDML, MadGraph, Pythia, CORSIKA 8; PyTorch, Pyro, Lightning, TorchScript/LibTorch, ONNX Runtime, SOFIE, LWTNN, TensorFlow.

References

Available on request.